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from flask import Flask, request, jsonify

import nltk

from nltk.chat.util import Chat, reflections

from textblob import TextBlob

from vaderSentiment.vaderSentiment import SentimentIntensityAnalyzer

from transformers import pipeline

app = Flask(__name__)

chatbot = pipeline("conversational", model="microsoft/DialoGPT-medium")

analyzer = SentimentIntensityAnalyzer()

responses = [

    (r'Hi|Hello|Hey', "Hello! How can I assist you today?"),

    (r'What is your name?', "I am an intelligent chatbot here to help you."),

    (r'Bye|Goodbye', "Goodbye! Have a great day!"),

    (r'I need help with (.*)', "Sure, I can help you with %1. Could you tell me
more?"),

    (r'(.*)', "I’m sorry, I didn’t quite understand that.")

]

chatbot_rules = Chat(responses, reflections)

def get_sentiment(text):

    blob = TextBlob(text)

    sentiment_score = analyzer.polarity_scores(text)

    if sentiment_score['compound'] >= 0.05:

        sentiment = 'positive'
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    elif sentiment_score['compound'] <= -0.05:
        sentiment = 'negative'
    else:
        sentiment = 'neutral'
    return sentiment, sentiment_score
def generate_response(user_input):
    sentiment, sentiment_score = get_sentiment(user_input)
    if sentiment == 'negative':
        response = "I'm sorry you're feeling that way. How can I assist you
further?"
    else:
        response = chatbot(user_input)


    return response, sentiment, sentiment_score
@app.route('/chat', methods=['POST'])
def chat():
    user_input = request.json.get('message')
    if not user_input:
        return jsonify({'error': 'No input message received.'}), 400
    response, sentiment, sentiment_score = generate_response(user_input)
    return jsonify({
        'response': response,
        'sentiment': sentiment,
```

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        'sentiment_score': sentiment_score
    })
if __name__ == "__main__":
    nltk.download('punkt')  # Download necessary NLTK resources
    app.run(debug=True)
```